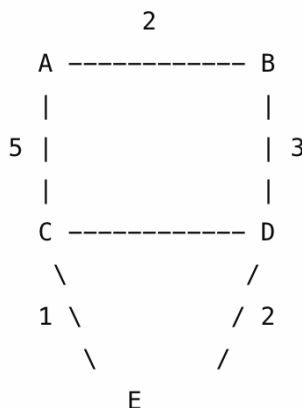


Exercise 2: Link State and Distance Vector Routing

Name: Sukanit Ritsatheanwong Student ID: 6710451437 Date: _____

Learning objectives: Apply Dijkstra and Bellman-Ford; construct forwarding tables; analyze convergence, failures, and count-to-infinity.

Network topology:



Part 1 — Link State Routing

Question 1 — Dijkstra's Algorithm

Starting at router A, execute Dijkstra's algorithm. Show every update and the node confirmed at each step.

Step	Confirmed Node	D(B), Prev.	D(C), Prev.	D(D), Prev.	D(E), Prev.
0	A	2, A	5, A	∞ , -	∞ , -
1	B	2, A	5, A	5, B	∞ , -
2	C	2, A	3, A	5, B	6, C
3	D	2, A	3, A	3, B	6, C
4	E	2, A	3, A	3, B	6, C

Question 2 — Shortest-Path Tree

Draw the shortest-path tree rooted at A, then list the shortest path and total cost.

SPT sketch:

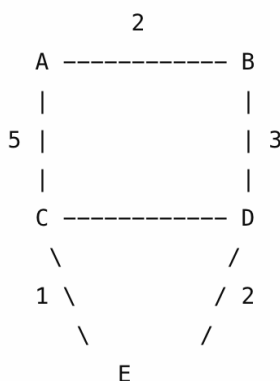
Destination	Shortest Path	Total Cost
B	A → B	2
C	A → C	5
D	A → B → D	5
E	A → C → E	6

Question 3 — Forwarding Table

Destination	Shortest Cost	Next Hop	Complete Path
B	2	B	A → B
C	5	C	A → C
D	5	B	A → B → D
E	6	C	A → C → E

Part 2 — Distance Vector Routing

Consider the same network. Initially, each router knows only the cost to its directly connected neighbors.



Question 4 — Bellman-Ford Update

Router A receives the following distance vectors from its neighbors:

From B:

A=2, B=0, C=4, D=3, E=5;

From C:

A=5, B=4, C=0, D=1, E=1.

Use the Bellman-Ford equation. Calculate router A's new distance to:

Show all calculations, including the neighbor through which the minimum-cost route is obtained.

Destination	Via B calculation	Via C calculation	Minimum Cost	Next Hop
B	$2 + 0 = 2$	$5 + 4 = 9$	2	B
C	$2 + 4 = 6$	$5 + 0 = 5$	5	C
D	$2 + 3 = 5$	$5 + 1 = 6$	5	B
E	$2 + 5 = 7$	$5 + 1 = 6$	6	C

Question 5 — Distance Vector Table

After receiving the distance vectors, construct router A's updated routing table.

Destination	Next Hop	Distance	Full Path
B	B	2	A → B
C	C	5	A → C
D	B	5	A → B → D
E	C	6	A → C → E

Part 3 — Routing Convergence

Question 6 — Link Failure

The C–D link fails. Explain separately what happens with Link State and Distance Vector routing.

Link State routing.

Your answer should describe:

- Which router(s) detect the failure.
- What information is flooded through the network.
- How routers update their link-state databases.
- How routers calculate new shortest paths.

Distance Vector routing.

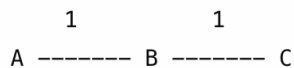
Your answer should describe:

- How routers learn about the failure.
- How updated distance vectors propagate.
- Why routers may temporarily have incorrect routing information.
- How routing eventually converges.

Link State	Distance Vector

Part 4 — Count-to-Infinity

Consider



Initially A reaches C through B at cost 2. The B–C link fails, but B receives stale information from A saying C is reachable at cost 2.

Question 7

1. Why can this create a routing loop?
2. Show how the estimated distance to C can increase over successive iterations.
3. What is count-to-infinity?
4. Why does it occur in Distance Vector routing?
5. Name a mitigation approach.

Part 5 — Compare LS and DV

Question 8

Question 6 — Link Failure

The C-D link fails. Explain separately what happens with Link State and Distance Vector routing.

Link State routing.

Your answer should describe:

- Which router(s) detect the failure.
- What information is flooded through the network.
- How routers update their link-state databases.
- How routers calculate new shortest paths.

Distance Vector routing.

Your answer should describe:

- How routers learn about the failure.
- How updated distance vectors propagate.
- Why routers may temporarily have incorrect routing information.
- How routing eventually converges.

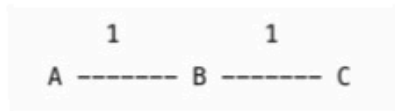
Link State Routing

- C-D link fail แล้ว Router C,D จะพบ link error แล้วจะส่ง Link-State Advertisement ให้ flooded ไปยัง router อื่นๆ โดย router จะอัปเดต Link-state Database แล้วคำนวณ shortest paths ใหม่ โดยไม่ใช้ C-D

Distance Vector Routing

- Router จะรับ distant vector จาก neighbor แล้วจะเปรียบเทียบว่า route ที่ได้จาก router นั้นเป็น distance และ next hop ของ router นั้นหรือไม่ ถ้าใช่จะรับค่า loop แล้วจะส่งค่า loop ไปยัง router นั้นๆ แล้ว non converge routing table ใหม่

Consider



Initially A reaches C through B at cost 2. The B–C link fails, but B receives stale information from A saying C is reachable at cost 2.

Question 7

1. Why can this create a routing loop?
2. Show how the estimated distance to C can increase over successive iterations.
3. What is count-to-infinity?
4. Why does it occur in Distance Vector routing?
5. Name a mitigation approach.

- 1.) A ส่ง ข้อมูลให้ B ว่าสามารถไปถึง C ได้โดยมี cost 2 แต่ B→C link fail B จึงส่ง ให้ A and so on
- 2.) จากข้อ 1.) เกิด ลูป/ซ้ำเรื่อย ๆ $A \rightarrow B \rightarrow C = 2$, $A \rightarrow B \rightarrow A \rightarrow C = 3$, $A \rightarrow B \rightarrow A \rightarrow B \rightarrow C = 4, \dots$
- 3.) router ใน distant vector ต้องมี metric เช่นการที่ ใช้เวลานานได้ จนเกิดเป็น protocol timeout infinity
- 4.) เพราะ distant vector รู้จาก neighbor ไม่รู้ค่าที่แน่นอนจาก topology
- 5.) Poisoned Reversed

Characteristic	Link State	Distance Vector
Basic algorithm	Dijkstra	Bellman-Ford
Information maintained	Link State Database/Topology Information	Distance Vector
Information exchanged	Link-State Information	Distance Vector กับ Neighbor
Algorithm	Shortest Path First	Bellman Ford
Topology knowledge	มี topology	รู้แค่ Neighbor
Routing updates	Flood link-state information	บอกกับ neighbor
Convergence	เร็ว	ช้า
Routing loops	ไม่มี routing loop topology	มี routing loop
Count-to-infinity	ไม่มี count-to-infinity	มี count-to-infinity
Example protocol	OSPF	RIP

Part 6 — Design Question

Question 9

You are designing a routing system for an enterprise with 500 routers requiring fast convergence, accurate topology knowledge, efficient routing, and hierarchical design. Choose LS or DV and justify your choice with at least three technical arguments. Your answer should discuss at least three advantages and/or disadvantages of your chosen approach.

Link State Routing

- Link State มี topology knowledge สามารถหา shortest paths ได้ ใน topology
- link state มี convergence ที่เร็วกว่า distance vector routing เพราะสามารถหา shortest paths ได้
- link state สามารถหา routing loop ได้ เพราะมี topology knowledge ในมือหา shortest paths ได้

Challenge — Algorithmic Analysis

Question 10

For N routers and E links, compare LS and DV in information maintained, communication overhead, computation, convergence, and scalability. Explain why large-scale intra-domain routing generally favors Link State.

	LS	DV	ข้อ domain มี topology information
information maintained	Link State Database/Topology Information	Distance Vector ของ Destination	แต่ละ router มี topology information ของ topology ใกล้เคียง
Communication overhead	Flood link-state information	บอก vector กับ neighbor	
Computation	ใช้ Dijkstra/SPF	ใช้ Bellman-Ford	
Convergence	เร็ว	ช้า	
Scalability	เหมาะกับ network ขนาดใหญ่ hierarchical design ได้	เหมาะกับ network ขนาดเล็ก	
Topology knowledge	มี topology knowledge	รู้แค่ neighbor topology	
Failure handling	Flood topology เมื่อมีปัญหา	บอกกับ neighbor	